

INTRODUCTION TO RESEARCH IN FINANCE

BE 451 Degree Project in Finance

Fall 2026

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TODAY'S AGENDA

- Timeline
- Synopsis
- Research Process
- Tutors
- Research Design with Data
- Questions

TIME-LINE & IMPORTANT DATES

- **September 7, 13.15 – 15.00:** Library workshop
- **September 3, 8, 15, 22:** Econometrics tutorial with Morteza (location: see Canvas, Portal)
- **September 8, 11, 15:** Database workshops
- **September 9, 13.00:** Deadline to submit the synopsis
- **September-November:** Individual meetings with tutors
- **November (date TBC):** Group meeting with tutor
- **December 7, 10.00am:** Submission deadline for thesis (via Canvas)
- **Before December 12:** Presentations and discussions
- **December 18, 18.00:** Deadline to submit via the “Publish Thesis” portal

DESIRED COURSE OUTCOME

Goal: Use state-of-the-art theory and empirical methods to analyze research questions in Finance & present the analysis

Criteria (not exhaustive)

- Useful: Is the thesis useful to a researcher in the relevant field?
- Correct: the analysis is accurate, use of proper methodology
- Relevant: fits the (recent) literature
- Contribution: does it extend existing research in a meaningful way?
- Independence

DESIRED COURSE OUTCOME

Important

- Writing a thesis can be a unique and immensely rewarding experience
- Embrace uncertainty and answer a question to which there are no predefined answers
- Answering questions "too big" cannot be done in 3 months (a research paper typically takes at the minimum 3 years from start to publication in a journal)
- Do not chase "statistical significance". What is important is that you have interesting and reasonable hypotheses
- A good thesis does not only shed light on the "what" (e.g., do ESG-focused investment mandates affect fund returns) but also "why" (which precise channels are at play?)

WHAT IS THE SYNOPSIS?

Broadly defines **two possible thesis topics** of interest

- Read document “Synopsis” on Canvas
- Read document “Tutors and topics” on Canvas

1 - 2 pages per topic

1. Clearly defined topic and quantitative exercise
2. Only one of the topics can be based on suggestions by tutors
3. Show that your question is *feasible*: you have access to the required datasets, time constraints
4. Short literature review: 2-3 papers (“top” journals – see document “Synopsis”) closely related to your research question and explain how you build on them and extend their contribution
5. *Do not copy / modify topics from previous theses, start with the academic literature, not prior theses*

SUBMITTING THE SYNOPSIS

- The synopsis should be submitted on Canvas as a .pdf file.
- **File title:** *BScThesis Surname1 Surname2.pdf*
- Students work in pairs.
- You can suggest tutors in your synopsis document.
- *Note that **cross-department theses are not supported**.*
 - Meaning: if you want to have the grade for a thesis in Finance on your transcript, you must write a thesis in Finance (supervised by a tutor appointed by the Department of Finance) and team up with a student in Finance.

AFTER THE SYNOPSIS

Following the submission of the synopsis, you will be assigned a tutor.

- Do not contact tutors yourself before being assigned your tutor.
- Students writing the synopsis alone will be assigned a partner.
- The allocation of tutors will occur within a week of the submission of the synopsis.

QUANTITATIVE THESES

- Most BSc theses in Finance are *quantitative*: model calibration, regressions
- It is very difficult to produce insightful *theory* papers in 3 months.
- Writing a good *case study* is also very challenging in the time-frame available, but you may suggest a qualitative case study for *one* of your two synopses
- Note: It is hard to make a clear contribution to the literature with a qualitative thesis

REPLICATION AND EXTENSION OF A PUBLISHED PAPER

- Standard for BSc thesis: replication *and* extension of a (recent) published paper in a top journal
- Guarantees that thesis is at frontier of research, uses correct methodology, asking relevant question
- The extension needs to be *meaningful*
 - Avoid default option of applying the question to Nordic countries unless meaningful
 - Potential extensions: different dataset, time, application, methodology, same method but different question, etc. (usually a meaningful extension will fulfill more than one of these criteria)
- Departing from the suggested format (extension of a paper) is risky. It can work if the question is very original and interesting
 - But: time-constraint
 - Potential high reward, but very high risk

FINDING A TOPIC

- Tutors provide better guidance if you work on their area of expertise
- Look at tutors' websites, research papers for relevant topics
- Suggested topics in the “Tutors and topics” document
 - These suggestions are often “raw” and are not a guarantee for successful research question!
- Read recent papers (<15 years) in top finance journals
 - Read only articles from top journals in Finance and Economics or working papers of researchers in top universities for recent topics
 - Complete list of top journals in the Syllabus; do not read other articles or other BSc theses!
 - *Once you have identified a topic and question, perform a full literature review (including relevant theses) and acknowledge relevant work in your thesis*

FINDING A TOPIC

- Academic analysis of recent interesting events?
- If you already have a precise question in mind, try to find existing papers that are the closest to your idea.
- Use Google Scholar to navigate the literature
 - Read in detail only articles published in top journals in Finance and Economics!
 - Check most cited articles (>100 citations unless papers are recent)

FEASIBILITY

- Many questions are interesting, few are feasible!
- Do you have access to the data?
 - Use the WRDS account (see Canvas for instructions)
 - Use resources made available by the library (go to the library workshops!)
- Can you learn and apply the methodology?
 - Simpler is better!
 - Attend Morteza's econometrics seminars
 - If you cannot justify why you use a complicated method, do not use it
 - Do not overestimate what can be done in 3 months
 - Get started *NOW*

SUBSTANTIVE REQUIREMENTS

Goal: The thesis should be as close as possible, in all dimensions, to a published article in a top journal of finance (list: Syllabus).

- This is an *aspirational* goal. We are aware of your time, data, and training constraints
- Question based on solid theoretical and/or empirical grounds
- Transparent formulation of the research question
- Generation of new knowledge and understanding of current state of knowledge
- Make explicit use of relevant theory and/or models
- Robust methods of data collection and analysis
- Follow the structure and writing style of articles from top journals
 - Examples: NO to tables of contents; YES to table captions that are detailed and contain all the relevant information to understand the table

SUPPORT

- Tutor: research questions
- TA (Morteza Aghajanzadeh): questions about coding and econometrics
- Course director (Riccardo Sabbatucci): any other question related to the thesis or if you have major issues with the process
- Important
 - Read the documents on Canvas
 - Follow instructions
 - Do **NOT** miss deadlines

MEETING WITH TUTORS

- Mission of the tutors: help you pass the thesis requirement
 - Anything beyond that is your responsibility, i.e., they are not required to nor may not be able to “guarantee” that the thesis meets certain quality standards above minimum ones
- Do not discuss grades with your tutor!
 - Any question not directly related to research should be directed to me
 - I grade the theses, not the tutors
- Come to meetings with a plan
 - Work, prepare ahead of the meeting or it will be a waste of precious time
 - Bring raw material that the tutors can work with
 - Should we do A or B? not: what should we do?
- Listen carefully, take notes, and act on the comments
- Some groups will have fewer / more meetings than other groups
 - Depends on topic, strength of group, time available for tutoring (tutors are busy) etc.
 - Tutors provide help but are not service providers
 - Have reasonable expectations about the tutoring process

RESEARCH PROCESS

1. Explore the frontier of the literature

- Find the closest, most relevant papers
- Check with your tutor to see if you understand the literature

2. Define a quantitative methodology

- What is the methodology used in the literature?
- Can you obtain the relevant data to answer your question?
- What is the simplest way to identify the mechanism studied?
- Build your contribution on solid ground by replicating the main quantitative exercise of a key paper in the literature

3. Starting writing early will help you formulate your questions

- Constant feedback between writing and research

WRITING

- Read published papers in top journal (see list in the Syllabus) to understand how to write a paper in Finance/Economics
- Do not write it as a student project with a table of contents, a complete literature review, off-topic digressions, and pedagogic explanation of standard methodology (e.g., OLS) and background (do not do that!)
- An *academic paper* is straightforward/to the point: every section, paragraph, sentence has a clear objective and is necessary
- An *academic paper* is addressed to someone knowledgeable about the topic, not to BSc students
- Familiarize yourself with scientific writing by reading articles in top journals and attempt to emulate as much as possible this writing style and format in your BSc thesis
- The formatting of the final thesis (e.g., tables, figures, references) should look like a paper in one of the top 3 finance journals (JF, RFS, JFE)

WRITING

As close as possible to a published article in a top journal in Finance

- Introduction (2-4 pages)
 - Paragraphs: background/motivation, research question, methodology, results, interpretation
 - What is the question? Why is it interesting/relevant? What is the methodology? What are the results?
 - Intro = self contained summary of paper
- Related Literature (<1 page)
 - What are the *closest* papers and how does the thesis differ?
 - What is the *contribution* to the literature?
 - It is *not* a review of the literature!
- Data and Methodology
- Main Empirical Analysis
- Interpretation of Empirical Findings / Discussion
- Conclusion
- Total length: up to 40 pages, single-spaced equivalent (including tables)

RESEARCH DESIGN WITH DATA

Textbook procedure for an empirical project:

1. Ask an important / interesting research question
2. Establish a credible estimation strategy
3. Use relevant data for the question

In Finance, most of the data is observational instead of experimental

- Sample selection bias
- Minimum detectable effect: check the power of your test

THE PROBLEM OF CAUSALITY

Consider the following questions:

- Does financial distress reduce enterprise value?
- Do high cash flows induce firms to invest more?
- Do CEOs recruited externally create less value?

Correlations little informative because of:

- Confounding factors (omitted variables)
- Reverse causality

Answers to these questions require *identification* of causal links

SOLUTIONS

The **Statistical/Experimental/Reduced form** approach

- No assumption made on firms' behavior
- Counterfactual identified for firms whose treatment status is as good as random
- Appropriate for corporate finance issues

The **Structural approach**

- Behavior of independent variables modelled using theory
- Counterfactual identified when reality behaves like model
- Appropriate for asset pricing

REGRESSION

Why the Regression?

- Useful for both prediction and identification (of causal links)
- Very robust (does a good job in wide variety of contexts)
- Starting base for most advanced empirical methods
- Very transparent (easy to replicate)

IMPORTANT REMARKS

- The thesis requires a lot of independent work (more than 15 credits worth)
- Do not procrastinate, begin *today*.
 - Most thesis groups that start too late run into troubles
- The research process can be frustrating. The feeling of being lost is part of the process as by definition you are pushing the research frontier.
- A *robust* research question is a question that is interesting independently of the results
 - Interest of the question should not depend on whether you confirm a hypothesis
- A good research question builds on strong foundations
 - Impossible to design a relevant research question without building on other papers

IMPORTANT REMARKS

Plagiarism

- We will perform a plagiarism check on the thesis
- You need to cite **all** your sources
- You will be in trouble if you copy from an existing thesis
 - For example, copy passages without citing
 - Copy table formats without citing
 - Copy code without citing
- **YOU NEED TO CITE ALL YOUR SOURCES, INCLUDING OTHER THESES!**

COMMON ISSUES

- It is difficult, coding is hard, data is challenging
 - Yes, this is most likely the most challenging (but also rewarding!) quantitative work you will do during your BSc. AI can help with coding tasks.
- Need more help, need more structure, more Q&As
 - Morteza can help you with econometrics and coding, your tutor can help you with research direction, and I can help you with anything else; contact us (asap, do not wait until problems are incurable)!
 - One core foundation of a thesis is to learn to conduct *independent* work
- I don't want to start with replication
 - If you can produce an original research question that is meaningful, please do so!
 - Replication is just the first step to finding something new & interesting, it is a guarantee of solid foundations such that we can trust your results
 - Hard to swallow pill: your original idea may not be as good as you think (possible, but: if it's so interesting, why hasn't it been done?)

QUESTIONS ?